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The Director

of the United States Patent and Trademark Office has received an application for a patent for a new and useful invention. The title and description of the invention are enclosed. The requirements of law have been complied with, and it has been determined that a patent on the invention shall be granted under the law.

Therefore, this United States

Patent

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(12) **United States Patent**
Ko et al.

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(54) **RADON SENSOR DEVICE USING POLYHEDRAL-SHAPED IONIZATION CHAMBER**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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CPC G01T 1/185; G01T 1/178; G01T 7/00
See application file for complete search history.

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(57) **ABSTRACT**

A radon sensor device using a polyhedral-shaped ionization chamber is proposed. The radon sensor device includes: an ionization chamber having an open side and inner sides surrounded by a first conductor and generating an electrical field therein by applying bias power to the first conductor; a cover having a first side covered with a second conductor and closing the open side of the ionization chamber such that that first conductor disposed on the inner sides of the ionization chamber and the second conductor are electrically connected; a probe unit disposed in the ionization chamber and absorbing ion charges produced when alpha (α) decay occurs in the ionization chamber; and a measurement circuit detecting an alpha particle detection signal by amplifying and processing an electrical micro-signal input from the probe unit into a predetermined magnitude.

19 Claims, 8 Drawing Sheets

